

Aligning the Czech Core Metadata Model with DataCite: A Real-World Matching Case Study

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Abstract

This poster paper presents complex mappings and bidirectional transformations between the Czech Core Metadata Model for Research Data (CCMM) and the DataCite Metadata Schema. The approach combines semantic- and XML-level complex mappings, implemented using XSLT and validated through round-tripping tests. The resulting mappings enable metadata exchange between national research data infrastructures and DataCite services, supporting interoperability and future RDF-based transformations. The poster paper is equipped with the GitHub repository at: <https://github.com/techlib/CCMM-DataCite-mappings>

Keywords

Metadata Schema Matching, Crosswalks, Semantic Interoperability, CCMM, DataCite

1. Introduction and Related Work

Matching heterogeneous metadata schemas is a foundational challenge in semantic interoperability. While ontology matching traditionally operates on RDF/OWL structures, many real-world systems still rely on XML-based schemas. Bridging these worlds through executable crosswalks presents a practical use case for complex schema matching.

This poster paper describes the background and methodology for designing CCMM (Czech Core Metadata Model for Research Data) and DataCite mappings and transformations. The motivation for these mappings and mutual transformations is twofold. On the one hand, it is necessary to export national CCMM metadata to DataCite-compliant metadata to obtain Digital Object Identifiers (DOIs). On the other hand, harvesting global DataCite records into the Czech National Metadata Archive (NMA) system requires backward conversion. For mappings and transformations a public GitHub repository¹ has been created to describe the entire mapping and XSLT transformation process and to provide implementations for both directions. Mappings and transformations consider CCMM version 1.1.0 and DataCite Metadata Schema version 4.7.

The Czech Core Metadata Model for Research Data is the primary metadata model for describing research data in Czechia. It is an application profile represented in the Data Specification Vocabulary (DSV) [2]. Compared with DCat-AP-Plus [6], it focuses on funding information,

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¹This release at <https://github.com/techlib/CCMM-DataCite-mappings/tree/release-ccmm-1.1.0-datacite-4.7-alpha1>

terms of use, geolocation, and tight liaison with controlled vocabularies such as location relations. Similarly to other nation-wide metadata schemas, CCMM is a unified schema for the (Czech) National Data Infrastructure, and it allows unified discovery in research data. DataCite is a global XML-based metadata standard for research output citation.

Mappings play an important role in the semantic interoperability of systems based on different semantic artefacts (e.g., ontologies). Ontology matching is already an advanced field with a relatively large number of (semi-)automatic approaches. Ontology matching has also been advanced by the Ontology Alignment Evaluation Initiative (OAEI), which has brought together providers of ontology matching test cases and ontology matching tool developers since 2004. Although most OAEI tracks focused primarily on various types of ontologies, in 2022, the Crosswalks Data Schema Matching track aimed to match 15 research metadata schemas, such as Dublin Core, Data Catalogue Vocabulary (DCAT), Data Catalogue Vocabulary - Application Profile (DCAT-AP), and DataCite Schema to Schema.org [5, 7]. The track was based on the reference alignment produced by the Research Data Alliance (RDA). In 2021, several mappings between metadata schemas have been created [4], such as DataCite to Dublin Core. Existing mappings are mostly represented as tables, and there are no ready-to-use, actionable formats such as XSLT. Moreover, CCMM has not been covered in mappings yet.

2. An Iterative Approach and Round-Tripping

The initial, straightforward mapping approach was based on a high-level (vocabulary-level) comparison of the metadata elements defined in CCMM 1.1.0 and in DataCite’s in-house OWL vocabulary representation. Several CCMM entities are profiled from DataCite. This initial straightforward approach (i.e., equivalent-based matching) takes CCMM entities and their DataCite-profiled counterparts as anchors. Next, XML mappings were based on previous anchors and also on sample examples of CCMM-compliant metadata and DataCite-compliant metadata. Subsequently, an exhaustive XML mapping has been applied on the level of XPath. Finally, XML mappings have been operationalized using XSLT transformations, aiming to transform CCMM and DataCite metadata in both directions.

XML mappings and operationalized XSLT transformations were created iteratively, driven by metadata examples. Initially, the approach has been based on four existing CCMM-compliant metadata.² For validation of the XSLT transformation, we employed round-tripping tests, ensuring that the transformed output can be converted back to its original form. The resulting metadata are syntactically valid and semantically consistent. However, due to the datacite semantic bottleneck (as described in Section 3), the resulting metadata has lost some information after the round trip. Furthermore, controlled vocabulary alignments were formalized using the SSSOM standard [3], contributing a clean mapping set to the community.³

Our workflow, combining high-level RDF alignment, XML mapping, XSLT operationalization, and round-tripping, provides a reusable framework for local-to-global schema alignment.

²<https://github.com/techlib/CCMM-DataCite-mappings#41-ccmm-xml-samples>

³For example, we prepared mappings between DataCite and CCMM regarding resource types at <https://github.com/techlib/CCMM-DataCite-mappings/blob/dev/exhaustive-mapping/mappings/mapping-datacite-coar-resource-types.csv>

3. Mapping Types and Metadata Schemas Distinctions

The alignment required addressing both simple and complex structural matches. Simple mappings (e.g., ‘datacite:awardTitle’ = ‘ccmm:award_title’) and value transformations (e.g., tokenizing strings or formatting dates) were straightforward. However, complex conditional mappings were also prominent. For example, ‘datacite:resource/contributors/contributor’ was derived from ‘ccmm:qualified_relation’, but only for relations whose ‘role’ was neither ‘Creator’ nor ‘Publisher’, as these roles were mapped to dedicated DataCite elements.

Testing the transformations on a random sample of 15 DataCite records revealed systemic schema distinctions that complicate automatic matching: (i) **Syntactic Heterogeneity**: Differences at the XML representations level, such as distinct element nesting, attribute usage, and string/date formatting requirements. These are directly operationalized using XPath expressions and XSLT functions. (ii) **Structural heterogeneity**: DataCite utilizes explicit, separate elements (creator, contributor, publisher), while CCMM employs a unified structure of qualified relations bound to an Agent Role vocabulary. Our approach handles this via conditional XSLT rules based on role filtering. (iii) **Semantic Heterogeneity**: DataCite uses free-text fields for rights, whereas CCMM strictly requires IRIs from controlled vocabularies and separates licenses from access rights. Furthermore, CCMM-specific concepts (e.g., explicit distribution formats or data management roles) lack direct structural equivalents in DataCite and must be derived or approximated using context-aware heuristics.

4. Conclusion and Future Work

This poster paper presents our statement of interest describing the background and methodology for designing CCMM (Czech Core Metadata Model for Research Data) and DataCite mappings and transformations. Fully automated matching tools struggle with complex 1 : N conditional rules and structural transformations (e.g., flattened vs. nested agent roles). While 1:1 candidate anchors can be automatically discovered, expert human-in-the-loop engineering remains indispensable for operationalizing executable crosswalks. While the mapping is currently maintained between XSD schemas, we are working on a complete high-level mapping using semantic standards such as RDFS and OWL, with semi-automated support. Similarly, the operationalization is currently performed using an XSLT transformation, while we are working on a SPARQL CONSTRUCT-based analogy for RDF metadata representation. All in all, a transition from the XML realm to the RDF world is underway. The work in progress could also be arranged for the next OAEI track, which focuses on metadata schema matching.

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